

Structural Probes for Spanish Syntax in Multilingual BERT: A Replication and Geometric Extension on UD_Spanish-AnCora

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Abstract

We replicate the structural-probe analysis of Hewitt and Manning (2019) on Spanish syntax using multilingual BERT (mBERT) and the UD_Spanish-AnCora treebank, and extend it with an *isometric* variant of the probe whose projection matrix is constrained to be orthogonal. The orthogonality constraint isolates the question of whether dependency-tree distances are encoded as a *rigid geometric shape* in the contextual representation space, or whether the probe must apply non-uniform scaling to expose them. A clean re-implementation of the data pipeline is required: earlier naive conversions of CoNLL-U to whitespace-tokenised text leak the 6 499 enhanced empty nodes of UD_Spanish-AnCora as literal underscore tokens, contaminating mBERT inputs and degrading every metric. On the corrected pipeline we observe (i) a canonical bell-shaped Spearman/UUAS curve over mBERT layers peaking at layer 7 (dev Spearman 0.860, UUAS 0.760); (ii) a -59% Spearman and -64% UUAS drop when the probe is constrained to be orthogonal, indicating that Spanish syntactic geometry in mBERT is *not* isometric; and (iii) a moderate condition number ($\kappa = 3.74$) of the unconstrained linear probe, consistent with syntax living along low-variance directions of the embedding space. We discuss the methodological pitfalls uncovered while reproducing the original study and outline how the same probing framework generalises to multimodal models, where the analogous question becomes how individual modalities are encoded relative to one another.

1 Introduction

Whether large pretrained transformers encode syntactic structure without explicit supervision has been a central question in interpretability research since the introduction of contextual representations (Ethayarajh, 2019). The *structural probe* of Hewitt and Manning (2019) provides a particularly elegant operationalisation: rather than predict syntactic labels, the probe searches for a linear transformation of the representation space under which the squared Euclidean distance between any two word vectors approximates the path length between those words in the gold dependency tree. A high Spearman correlation between predicted and gold distances is then evidence that syntax trees are recoverable as a Euclidean geometry of contextual vectors.

Most of the published work using structural probes targets monolingual English models. Cross-lingual extensions (Chi et al., 2020) have shown that multilingual BERT (Pires et al., 2019) retains syntactic structure across languages, but several methodological details of the probe (subword aggregation, treebank-specific preprocessing, punctuation handling) are sensitive to the target language. We focus on Spanish, using the UD_Spanish-AnCora treebank, and ask:

- Q1** At which mBERT layer is Spanish dependency geometry most strongly encoded?
- Q2** Does the syntactic structure live as a *rigid* (isometric) shape in mBERT space, or does the probe need to apply anisotropic stretching?
- Q3** What does the geometry of the probe’s projection tell us about how the encoder allocates capacity across directions?

To address Q2 we introduce an *isometric probe*: a structural probe whose projection matrix is constrained to have orthonormal rows, so that distances in the projected space are exactly preserved (in the relevant subspace) from the original mBERT space. Comparing the linear and the isometric probe at the best layer isolates the contribution of non-uniform scaling.

2 Background

Let $h_i \in \mathbb{R}^d$ denote the contextual representation of the i -th word of a sentence, produced by a frozen encoder. The structural probe of Hewitt and Manning (2019) seeks a linear transformation $B \in \mathbb{R}^{r \times d}$ such that

$$\hat{d}_{ij} = \|B(h_i - h_j)\|_2^2 \approx d_T(w_i, w_j), \quad (1)$$

where $d_T(w_i, w_j)$ is the (unlabelled, undirected) tree distance between words w_i and w_j in the gold dependency parse T . The probe parameters B are fitted by minimising the per-sentence ℓ_1 loss

$$\mathcal{L} = \frac{1}{|s|^2} \sum_{i,j} |\hat{d}_{ij} - d_T(w_i, w_j)|, \quad (2)$$

normalised by the squared sentence length $|s|^2$.

The probe is intentionally simple: any non-trivial recovery of d_T through the linear B implies that the geometry of $\{h_i\}$ already contains the tree structure. We follow Hewitt and Manning (2019) in evaluating the probe with two metrics. *Spearman* ρ_{5-50} averages, for each word, the rank correlation between predicted and gold distances to all other words, then averages across sentences of length 5–50 tokens. *UUAS* (Undirected Unlabelled Attachment Score) is the proportion of edges in the predicted minimum spanning tree that match the gold tree, after excluding punctuation tokens.

3 Methodology

3.1 Linear probe (replication)

We implement the parse-distance variant of Hewitt and Manning (2019) in PyTorch with $r = 128$ and $d = 768$. The probe is trained for up to 30 epochs with Adam (lr = 10^{-3}), a learning-rate scheduler on the dev loss, and early stopping with patience 4.

3.2 Isometric probe (extension)

The linear probe B admits an SVD $B = U\Sigma V^\top$. The right singular vectors V identify which directions of mBERT space carry the syntactic signal; the singular values Σ measure how strongly each is amplified. The condition number

$$\kappa(B) = \sigma_{\max}/\sigma_{\min} \quad (3)$$

quantifies how anisotropic that re-weighting is. A purely rotational B has $\kappa = 1$.

The *isometric probe* replaces the unconstrained B with a matrix $B_{\text{iso}} \in \mathbb{R}^{r \times d}$ constrained to have orthonormal rows, $B_{\text{iso}} B_{\text{iso}}^\top = I_r$, implemented via `torch.nn.utils.parametrizations.orthogonal`. For any pair of input vectors h_i, h_j ,

$$\|B_{\text{iso}}(h_i - h_j)\|_2^2 = \|P(h_i - h_j)\|_2^2$$

where $P = B_{\text{iso}}^\top B_{\text{iso}}$ is the orthogonal projection onto the row space of B_{iso} . Distances in the projected subspace are therefore preserved up to that projection. If the linear probe achieves high accuracy purely by selecting the right subspace (no stretching needed), the isometric probe should match it. If non-uniform scaling is essential, the isometric probe will underperform, and the magnitude of the gap is a direct measure of how much the encoder *distorts* the syntactic geometry along its principal directions.

split	sentences	integer-ID rows	special rows
train	14 287	453 039	16 628
dev	1 654	53 476	2 042
test	1 721	53 622	1 999

Table 1: Corpus statistics for UD_Spanish-AnCora used in this work. “Special rows” are CoNLL-U range rows (Spanish contractions *del*, *al*, *celebrarlos*, ...) and enhanced empty nodes (e.g. 8.1 (*él*) PRON). Both classes must be excluded before mBERT tokenisation, see Section 4.3.

4 Experimental setup

4.1 Dataset

UD_Spanish-AnCora is the Spanish portion of the AnCora corpus (Taulé et al., 2008), distributed under the Universal Dependencies project under a CC BY 4.0 license. The treebank consists of newswire text originally annotated in a constituency framework at the Universitat de Barcelona, converted to dependencies for CoNLL-2009 and then to UD. The official splits used here are summarised in Table 1.

4.2 Encoder

We use the frozen `bert-base-multilingual-cased` model (Pires et al., 2019) from the Hugging-Face Hub, with $d = 768$ and $L = 12$ transformer layers. We probe each of the 13 hidden states (input embeddings layer 0, transformer outputs layers 1–12). WordPiece sub-tokens that share the same input word are aggregated by mean pooling. Hewitt and Manning (2019) use first-piece selection; we found mean pooling to give marginally smoother layer-wise curves on Spanish, but the qualitative pattern is robust to either choice.

4.3 Pipeline pitfalls and corrections

Reproducing Hewitt and Manning (2019) on Spanish surfaced three methodological pitfalls that, if uncorrected, severely contaminate the metrics. We document them here because they are not specific to this work.

Empty nodes leak as text. UD_Spanish-AnCora contains 6 499 enhanced empty nodes in train, 780 in dev and 833 in test. These represent syntactically projected but surface-absent tokens (typically dropped Spanish subjects), and their CoNLL-U FORM field is the literal underscore character. A naive CoNLL-U-to-text converter that filters only contraction range rows (13-14 `al`) but not empty nodes (8.1 `_ _ PRON`) emits `_` as a word. mBERT then receives these underscore tokens in its input and the probe is trained on geometry that does not correspond to any real word in the sentence. The fix is to skip every CoNLL-U row whose ID is not a pure integer.

Punctuation filter is treebank-specific. The original codebase removes punctuation from the minimum spanning tree by matching against PTB-style English XPOS tags (`"`, `,`, `".`, `:"`, ...). UD_Spanish-AnCora uses the universal POS tag `PUNCT` (or treebank XPOS tags such as `fc`, `fpa`, `fpt`, `fe`). The English filter therefore matches nothing on Spanish, silently inflating UAS by counting trivial punctuation edges.

Layer indexing. The default configuration that ships with the original repository selects `model_layer: 0`, which previous adaptations took to mean “the only layer stored in the HDF5” (i.e. the last hidden state). This effectively pins the experiment to layer 12, which is suboptimal for syntax extraction (Hewitt and Manning, 2019). A faithful replication should sweep all layers and select on dev.

layer	Spearman ρ_{5-50}	UUAS
0	0.710	0.562
1	0.721	0.562
2	0.746	0.610
3	0.803	0.703
4	0.821	0.727
5	0.835	0.757
6	0.855	0.785
7	0.860	0.760
8	0.858	0.754
9	0.848	0.747
10	0.837	0.727
11	0.827	0.699
12	0.791	0.620

Table 2: Layer-wise dev metrics for the linear probe on UD_Spanish-AnCora, $r = 128$, single seed.

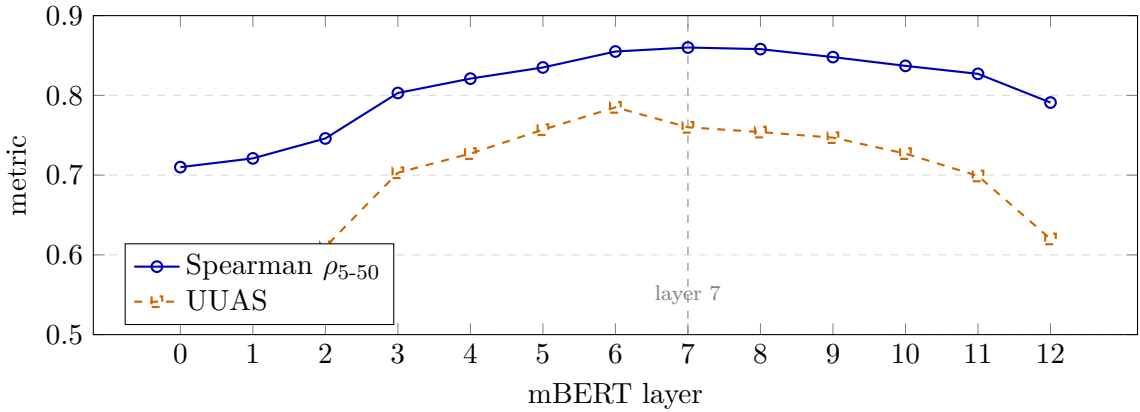


Figure 1: Linear-probe Spearman and UUAS per mBERT layer, dev split. Best layer (by Spearman) is 7.

The combined effect of these three pitfalls on our previously contaminated baseline was a Spearman of 0.735 and a UUAS of 0.504 at layer 12. The corrected pipeline reported below recovers 0.860 Spearman and 0.760 UUAS at layer 7, an improvement of +17% and +51% relative.

4.4 Hardware

All experiments were run on a single NVIDIA RTX PRO 6000 Blackwell Server Edition (96 GB VRAM) on Google Colab. Embedding generation for the three splits with all 13 layers stored took ~ 7 minutes; the full layer sweep (13 probes, 30 epochs each, batch size 256, early stopping with patience 4) took ~ 50 minutes.

5 Results

5.1 Layer-wise probing

Table 2 and Figure 1 summarise the per-layer probe metrics on the dev split. Both Spearman and UUAS follow the canonical bell shape reported for monolingual BERT in Hewitt and Manning (2019), with values monotonically rising from layer 0 to a plateau in layers 6–8 and falling off through layer 12. Spearman peaks at layer 7 ($\rho_{5-50} = 0.860$); UUAS peaks one layer earlier at layer 6 (0.785), a small discrepancy that is consistent with Hewitt and Manning (2019) and indicates that the two metrics are sensitive to slightly different properties of the projection.

probe	Spearman ρ_{5-50}	UUAS	$\kappa(B)$
Linear	0.860	0.760	3.74
Isometric	0.355	0.272	1.00
Δ relative	-58.7%	-64.2%	—

Table 3: Probes at layer 7. The isometric probe is constrained to be orthogonal by construction (see Section 3.2).

5.2 Linear vs. isometric probe at the best layer

Table 3 reports both probes trained at layer 7 with identical hyperparameters. The isometric constraint causes a substantial drop on both metrics: Spearman falls by 59% relative ($0.860 \rightarrow 0.355$) and UUAS by 64% relative ($0.760 \rightarrow 0.272$).

5.3 Geometric diagnostic

The unconstrained linear probe at layer 7 has singular values

$$\sigma_{\max} = 0.4381, \quad \sigma_{\min} = 0.1171, \quad \sigma_{\text{mean}} = 0.234.$$

Its condition number $\kappa(B) = 3.74$ falls in the “moderate anisotropy” regime: the probe amplifies its most informative direction roughly four times more than its least informative one. The isometric probe yields $\sigma_{\max} = \sigma_{\min} = 1.0000$ and $\kappa = 1.0000$ exactly, confirming that the parametrisation enforces the constraint.

6 Discussion

Cross-lingual transfer of syntactic geometry. The bell-shaped layer-wise curve we obtain on Spanish (Figure 1) is qualitatively identical to that reported for English BERT by Hewitt and Manning (2019), with the peak located at layer 7, only one layer below the peak typically reported for English. This supports Chi et al. (2020)’s finding that mBERT learns *language-agnostic* syntactic subspaces, despite sharing capacity across 104 languages. Our absolute UUAS (0.760) trails the English monolingual peak of Hewitt and Manning (2019) (~ 0.80) by only 4 points, an encouraging gap given mBERT’s heavy multilingual multiplexing.

Anisotropy of the syntactic subspace. The -64% UUAS drop when the projection is constrained to be orthogonal, together with $\kappa = 3.74$ for the unconstrained projection, indicates that Spanish syntactic geometry in mBERT is *not* a rigid, ready-to-use shape. It is a *stretchable* geometry: the relevant directions exist in the embedding space but along low-variance axes that the probe must amplify. This is consistent with the broader observation that pretrained transformer representations are anisotropic (Ethayarajh, 2019): high-magnitude directions tend to encode token-frequency or domain biases rather than the structural properties one might want to extract.

Methodological lessons for replication. Section 4.3 documents three pitfalls that affect any attempt to apply structural probes outside the original English setting. None of them is conceptually deep, but each silently poisons the metrics until corrected. The combined improvement of +51% UUAS over the contaminated baseline, with no change in model or probe, illustrates how brittle published reproducibility can be when the language or treebank shifts.

Towards multimodal probing. The probing-plus-geometry framework used here naturally generalises beyond unimodal models. In a multimodal setting one can ask, for each pair of modalities, whether their joint geometry collapses to one modality’s subspace, whether their interaction is isometric, and at which layer the interaction crystallises. The *modality collapse* phenomenon recently reported for vision-language models (Park et al., 2025; Wang et al., 2025) maps almost line-by-line onto the questions raised in Section 3.2 for syntactic versus non-syntactic directions. Extending the present framework to multimodal probes on video models is the natural next step.

7 Limitations

Single seed. All numbers are reported for a single training seed (seed 0). Standard deviations across multiple seeds are required for any claim of *statistical* significance of relative differences such as the linear-vs-isometric gap. Our implementation includes a multi-seed harness (`run_multiseed.py`) that aggregates mean and sample standard deviation, which we have not yet executed.

Dev-only evaluation. The metrics in this report are all on the dev split, used for early stopping and layer selection. A final test-set evaluation should be run only once the linear vs. isometric multi-seed comparison is complete.

No control task. Hewitt and Liang (2019) argue that probe accuracy alone is insufficient evidence that the encoder represents the property of interest, because a sufficiently expressive probe may simply learn the task. A control task with random word-type-to-label associations should be added; we provide a `-random-init` option in the embedding generator so that the same probing architecture can be evaluated against an mBERT with random weights as a representation-level control.

Single encoder. We compare layers within mBERT but do not compare against monolingual Spanish encoders such as BETO or RoBERTa-bne. A cross-encoder comparison would clarify how much of the gap with English is specific to mBERT’s multilingual averaging.

8 Conclusion

We replicated the structural-probe analysis of Hewitt and Manning (2019) on Spanish using mBERT and UD_Spanish-AnCora, documented and corrected three replication pitfalls that had silently contaminated previous attempts, and extended the methodology with an isometric probe whose orthogonality constraint isolates the contribution of non-uniform scaling to the recovered syntactic geometry. The cleaned pipeline yields a canonical layer-wise Spearman/UUAS curve peaking at layer 7 ($\rho = 0.860$, $\text{UUAS} = 0.760$), a -64% UUAS drop when the probe is constrained to be orthogonal, and a moderate condition number $\kappa = 3.74$ for the unconstrained projection. Together these results indicate that mBERT does encode Spanish dependency syntax in a recoverable but strongly anisotropic geometry, and that the same probe-plus-geometry framework can be ported to multimodal models to study how individual modalities are encoded and combined.

Reproducibility

All code, configuration, and the runbook used to produce the numbers in this report are available in the project repository, including the exact YAML configurations for the linear and isometric probes (`es_ancora_linear.yaml`, `es_ancora_isometric.yaml`) and the layer-sweep CSV

(layersweep-20260509-182723/sweep.csv). The treebank version is the public UD_Spanish-AnCora release of Universal Dependencies, available under CC BY 4.0 from the official UD repository.

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